

ACHINTYA RAI

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EDUCATION

University of California, San Diego

Bachelor of Science in Computer Engineering

Sept. 2024 – Dec. 2027

GPA: 3.9/4.0

- Competitive Programming Club, NSF REU Fellow (National Science Foundation), UC Scholars Research Fellow

WORK EXPERIENCE

Adobe

Software Engineering Intern

San Jose, CA

June 2026 – Sept. 2026

- Profiled **60.6M+** payment events in Databricks using Python, benchmarking data quality to flag **86,387** duplicate records and a **47%** null-timestamp rate.
- Isolated a **26-hour** outage across **1,272** benchmarked payment segments.
- Automated ML lifecycle retraining, evaluation, and hourly inference via SQL/MLOps pipelines.

Decklar (formerly Roambee)

Software Engineering Intern

Sunnyvale, CA

June 2024 – Aug. 2024

- Optimized Python/NumPy pipelines, boosting throughput **15%** for **1M+** weekly shipment events.
- Built visualization data structures for shipment dashboards, improving reporting speed **20%**.

VALT Health

Founding Software Engineer

Remote

Aug. 2025 – Feb. 2026

- Engineered a Flask backend and GCP workflows on Linux, unifying **100k+** datapoints across 20+ sources with version control.

RESEARCH

Graph Theory & Neural Architecture

Undergraduate Research Fellow (NSF REU)

San Diego, CA

March 2026 – Present

- Engineered a Python pipeline for deep learning model analysis, compressing **500GB** LLM checkpoints to **13GB** (Llama-2, DeepSeek-V2).
- Benchmarked and optimized placement scripts, achieving a **30x** runtime reduction on resource-constrained **8GB RAM** systems.
- Applied Louvain Clustering to raise model cluster density from **0.18%** to **13.25%** in Llama-2-70B.

Data Science Teaching & Learning Lab

Undergraduate Researcher

San Diego, CA

Sept. 2024 – Present

- Built an LLM data-labeling tool using **Python** and the **Gemini API**, automating labeling for **16k+** data points in the ML evaluation lifecycle.
- Reduced labeling time **90%** via iterative UI/UX refinements for high-throughput, human-in-the-loop verification.

PROJECTS

NanoGPT | Python, PyTorch, C++, Transformers, LLM Optimization | [🔗](#)

Mar. 2026

- Implemented a GPT-style transformer in **Python/PyTorch** with multi-headed attention layers.
- Built deep learning training/inference pipelines optimized in **C++**, cutting per-token latency **25%** and sustaining **100%** hardware utilization.

Watchdog | Python, Gemini API, Browser-Use, Asyncio, SQLite [🔗](#)

April 2026

- Built a concurrent **Python/Asyncio** CLI with non-blocking **aiosqlite** persistence.
- Optimized local state persistence for real-time task synchronization, reducing context-switching friction.

SquaredUpChess | React, TypeScript, Django, AWS, Docker [🔗](#)

Aug. 2025 – Dec. 2025

- Established a **CI/CD** pipeline on containerized **Docker/AWS** infra, ensuring **99.9%** deployment reliability.

TECHNICAL SKILLS

Languages: Python, C/C++, SQL (Postgres/SQLite), Java, Bash

Systems & Tools: Linux, version control, Docker, Databricks, MLOps, CI/CD, GCP, AWS

AI/ML & GPU Computing: PyTorch, CUDA, machine learning lifecycle, data visualization, dashboards